Document _ID

Unique ID: Har document ka _id alag hota hai. Same collection me 2 documents ka _id same nahi ho sakta.

Type: MongoDB me jo default _id banta hai uska type hota hai ObjectId.

Automatically Create Hota Hai: Agar aap insert karte time _id nahi dete, to MongoDB khud ek ObjectId generate kar deta hai.

Primary Key ki Tarah: SQL me jaise primary key hoti hai, MongoDB me _id waise hi behave karta hai.

Fast Search: _id par automatically index bana hota hai, isliye _id se search bahut fast hota hai.

Document Identification: Update, delete, ya specific document find karne ke liye _id use hota hai.

Immutable (Usually): Ek baar document create hone ke baad _id change nahi kiya ja sakta.

_Id Structure:

In MongoDB, an ObjectId is a 12-byte binary value. The reason you see it as 24 characters is because it is typically represented as a hexadecimal string

- 1 byte = 8 bits
- In Hexadecimal, 1 byte is represented by **2 hex characters**

_Id Breakdown:

1. Timestamp (First 4 bytes)
   - Ye batata hai document kab create hua.
   - ObjectId se approximate creation time pata chal sakta hai.
   - Example: agar ObjectId 65f1a2... hai to uska timestamp decode kiya ja sakta hai.
2. Machine Identifier (Next 3 bytes)
   - Ye identify karta hai kaunse machine/server ne document create kiya.
   - Distributed systems me helpful hota hai.
3. Process ID (Next 2 bytes)
   - Ye batata hai kaunsa process document generate kar raha tha.
4. Counter (Last 3 bytes)
   - Ye ek incrementing counter hota hai.
   - Same second me multiple documents insert hone par unko unique banata hai.

Thank You